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Department of Electronics and Communication Engineering -- KLVZA

Course Handout

2025-2026, Odd Sem

Course Title	:DIGITAL SYSTEM DESIGN
Course Code	:25EC1102
L-T-P-S Structure	: 3-0-2-0
Pre-requisite	:
Credits	: 4
Course Coordinator	:Lakshmi Kuruguntla
Team of Instructors	:
Teaching Associates	:

Syllabus :CO1: Fundamentals of Digital systems: Introduction to Digital systems, Number Systems, Signed number Representation, Binary codes, Code Conversion, Logic gates, Boolean Algebra and Minimization techniques: SOP & POS forms, Karnaugh maps (upto 4 variables). CO2:Combinational Logic Circuits: Design Procedure for combinational logic circuit, Adder and Subtractor, Encoders, Decoders, Multiplexers, De-multiplexers, Magnitude Comparators, Parity Generators and Checkers, BCD to seven segment decoder. CO3:Sequential Logic Circuits: Latches, Buffer, and Flip-Flops, Registers and Counters: Shift registers and types, Design of Synchronous and Asynchronous Counters, Modulus counters, Finite state machines: Moore and Mealy machines, Introduction to Threshold Logic, Sequence Generator, Sequence Detector. CO4:Programmable Logic Devices: Programmable Read only Memory (PROM), Programmable Array Logic (PAL), Programmable Logic Array (PLA), Designs using Programmable Devices. Complex Programmable Logic Devices, Field Programmable Gate Arrays, Applications of CPLDs and FPGAs.

Text Books :Text Books:1. M.Morris Mano, "Digital Logic and Computer Design", Pearson publication, 5th Edition – 2013 2. Stephen Brown, Zvonko Vranesic "Fundamentals of Digital Logic with Verilog Design" McGraw-Hill,, 3rd Edition. 3. Zvi Kohavi, Switching and Finite Automata Theory, McGraw-Hill computer science series,2nd Edition.

Reference Books :Reference Books:1. A Anand Kumar, "Digital Fundamentals", PHI publication, 4th edition-2016. 2.R.P. Jain, "Modern digital Electronics", Tata McGraw Hill, 4th edition-2009. 3. Ronald J. Tocci, Neal S. Widmer, Gregory L. Moss,"Digital Systems Principles and Applications", Prentice Hall, 10th edition-2014.

Web Links :[1]. <https://www.edx.org/learn/design/harvey-mudd-college-digital-design>. [2]. <https://www.youtube.com/playlist?list=PLBlnK6fEqRjMH3mWf6kwqITbT798eAOm>

MOOCS :[1]. <https://nptel.ac.in/courses/117106086> [2] <https://archive.nptel.ac.in/courses/106/105/106105163/> [3].<https://www.coursera.org/learn/digital-systems> [4]. https://onlinecourses.nptel.ac.in/noc25_ee125/preview

Course Rationale :The increasing demand for efficient and intelligent digital systems has made Digital System Design (DSD) a fundamental area in electronics and computer engineering. This course equips students with the theoretical foundations and practical skills needed to design, analyze, and simulate digital circuits. Emphasizing combinational and sequential logic, logic simplification techniques, and state machine design, the course enables students to develop a strong understanding of digital logic design. Through the use of simulation tools like Logisim and DSCH2, students gain hands-on experience in modeling and validating digital systems, preparing them for roles in embedded systems, digital design, and related domains.

Course Objectives :This course aims to introduce students to the fundamentals of Digital System Design (DSD), focusing on the analysis, design, and implementation of combinational and sequential logic circuits. It covers number systems, Boolean algebra, logic simplification techniques, and the use of Karnaugh maps for efficient circuit design. Emphasis is placed on the design of sequential components such as flip-flops, counters, and finite state machines, as well as the implementation of digital systems using Programmable Logic Devices (PLDs), including CPLDs and FPGAs. The course enables students to understand and apply digital logic principles in developing reliable and efficient embedded and real-time systems.

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Understand fundamental concepts of number systems, code conversions, Boolean algebra, and logic minimization.	PSO1	2
CO2	Design combinational logic circuits using logic minimization techniques	PSO1,PO1,PO2	3
CO3	Design sequential circuits using flip-flops, counters, and registers.	PSO1,PO1,PO2	3
CO4	Build logic functions using Programmable Logic Devices (PLDs).	PSO1,PO1,PO2	3
CO5	Develop and Analyze various digital logic circuits using combinational, sequential and FSM models for real time applications	PSO1,PO1,PO3,PO5	4

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-1	COI-2	COI-3	COI-4
CO1	2	Btl-1 Recall different types of number systems, binary codes, and basic Boolean laws.	Btl-2 Understand the logical expressions in different ways of representation. Recognize the optimization techniques to reduce circuit complexity		
CO2	3	Btl-1 List the types of basic combinational logic circuits such as adders, subtractors multiplexers, and decoders	Btl-2 Describe the method of reducing combinational logic circuits using Boolean algebra and Karnaugh maps for simplified implementation.	Btl-3 Design optimized combinational logic circuits using minimized Boolean expressions.	
CO3	3	Btl-1 Recall the characteristic tables and symbols of flip-flops used in sequential circuits.	Btl-2 Describe the functionality of flip-flops, counters, and registers, and their role in the design of sequential logic circuits.	Btl-3 Implement sequential circuits using flip-flops, counters, registers, and FSMs for timing and control applications such as sequence generation and detection.	
CO4	3	Btl-1 Recall the types and features of common Programmable Logic Devices such as PROMs, PALs, and PALs.	Btl-2 Describe the internal structure and configuration methods of PLDs used in logic circuit implementation.	Btl-3 Implement simplified logic functions using Programmable Logic Devices (PLDs) through appropriate design techniques.	
CO5	4	Btl-1 Recall the basic functions, symbols, and truth tables of combinational and sequential circuits including multiplexers, counters, shift registers, and programmable logic devices."	Btl-2 Describe the functionality and working principles of combinational and sequential digital circuits through application-based designs using multiplexers, counters, shift registers, programmable logic devices, and simulation environments.	Btl-3 "Implement combinational and sequential digital circuits for real-time applications using multiplexers, counters, shift registers, and programmable logic devices through simulation tools."	Btl-4 "Analyze the performance and logic behavior of application-specific combinational and sequential circuits designed using multiplexers, counters, shift registers, and programmable logic devices."

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome
PSO1	An ability to solve Electronics engineering problems, using latest hardware and software tools, to obtain appropriate solutions in the domain of embedded systems and Internet of things.
PO1	Engineering Knowledge:Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PO3	Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations
PO5	Modern Tool Usage:Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

Lecture Course DELIVERY Plan:

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
1	CO1	COI-1	Course Handout	Course Handout	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
2	CO1	COI-1	Introduction to Number Systems	R1, ch: 0.1-0.2, pp. 1	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
3	CO1	COI-2	Number Systems Conversions:(i) Decimal to Any Radix (Base r), (ii) Any Radix (Base r) to Decimal	R1, ch:1.1-1.5, pp. 29-65	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
4	CO1	COI-2	Signed number Representation and related basic operations	R1, ch:1.1-1.5, pp. 29-65	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
5	CO1	COI-2	Binary Codes	R1, ch:1.6-1.12,pp. 68-93	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
6	CO1	COI-2	Code Conversion	R1, ch:1.6-1.12, pp. 68-93	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
7	CO1	COI-2	Review of Boolean Algebra and De Morgan's Theorem	R1, ch:2.13, pp. 155-163	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
8	CO1	COI-2	Logic Gates	T1, ch: 1.9, pp. 30-31	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
9	CO1	COI-2	Minimization techniques: SOP, POS forms and Canonical representation	R1, ch:2.18,2.19,2.21,pp. 175-183	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
10	CO1	COI-2	Karnaugh Maps	R1, ch:3.1-3.4, pp. 231-248	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
11	CO1	COI-2	Karnaugh Maps with Don't cares	R1, ch:3.7,3.10, pp. 259-264, 269-272	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
12	CO2	COI-1	Design Procedure for combinational logic circuit	T1 ,CH4, Pageno129-130	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
13	CO2	COI-2	Adders and Subtractors	R1, ch:4.3, pp. 326-334	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
14	CO2	COI-2	Encoders, Decoders	R1, ch:4.4, pp. 374-388	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
15	CO2	COI-3	Applications towards logic circuit construction	R3, ch:4.20,4.22, pp. 374-388	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
16	CO2	COI-2	Multiplexers, De-multiplexers	R3, ch:4.24, pp. 389-397	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
17	CO2	COI-3	Applications towards logic circuits construction	R3, ch:4.25, pp. 389-397	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
18	CO2	COI-2	Magnitude comparator	R3, ch:4.18, pp. 370-374	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
19	CO2	COI-2	Parity Generators	R3, ch:4.17, pp. 366-369	Chalk,PPT,Talk	End Semester Exam,Home Assignment,SEM-EXAM1
20	CO2	COI-3	Parity Checkers	R3, ch:4.17, pp. 366-369	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
21	CO2	COI-3	BCD to seven segment decoder	R3, ch:4.23.8, pp. 386-388	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
22	CO3	COI-1	Introduction to Sequential Logic circuits and Memory elements	T1,CH 5,Pageno123- 126	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
23	CO3	COI-2	Latches & Buffer	T1 ,CH 5, Pageno186-190	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
24	CO3	COI-2	Flip-Flops and types: SR, JK, D and T (Truth table, Characteristic Table and Excitation Table).	T1,CH 6,Pageno 204-220	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
25	CO3	COI-3	Flip-Flop Conversions	T1,CH7,Page no 213-220	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
26	CO3	COI-2	Basic Concept of Register and Shift Registers	T1,CH7,Pageno 253	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
27	CO3	COI-3	Types of Shift Registers and related examples	T1,CH 7,Page no 258	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
28	CO3	COI-3	Counters and types: Asynchronous and Synchronous Counters	T1,CH7,Page no258-267	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
29	CO3	COI-3	Ring & Johnson Counter as Timing Control Units	T 1,CH7,Pageno 268-270	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
30	CO3	COI-3	Finite state machines: Moore and Mealy machines , Introduction to Threshold Logic	T[1],CH8,Page no305-308	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
31	CO3	COI-3	Sequence Generator, Sequence Detector	T1,CH12,Page no445	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
32	CO4	COI-2	Programmable Logic Devices (PLDs)	T1,CH 12,Page no446-448	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
33	CO4	COI-3	Programmable Read only Memory (PROM) and Designs using PROM	T2], CH12,Page no448-452	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
34	CO4	COI-3	Programmable Array Logic (PAL) and Designs using PAL	T2,CH12,Pageno 462-465	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
35	CO4	COI-3	Programmable Logic Array (PLA) and Designs using PLA	TBOOK[2],CH 12,Page no 466-469	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
36	CO4	COI-3	Complex Programmable Logic Devices (CPLD)	T2,CH 12,Page no469-476	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
37	CO4	COI-3	Field Programmable Gate Arrays (FPGA)	T2 ,CH 12,Pageno 478	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
38	CO4	COI-3	Applications of CPLDs and FPGAs	T2,CH 11,Page no392-396	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
39	CO4	COI-3	Revision on PLDs	NA	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2

Lecture Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Course introduction session

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
5	Attendance	2	Chalk	--- NOT APPLICABLE ---